Mohak Sharma

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Graduate researcher at Columbia working on neural interfaces for embodied agents: decoding noisy human intent (EEG, gaze) and pairing it with agent-side body models to drive precise, high-DOF action in real time.

RESEARCH INTERESTS

Neural interfaces for embodied agents: decoding human intent from non-invasive physiological signal (EEG, gaze), and the agent-side representations, particularly learned models of an agent's own body, that make sparse intent sufficient to drive high-DOF action. Also: shared autonomy, physics-informed neural fields, neural encoding models.

EDUCATION

Columbia University, M.S. Computer Science — Expected Dec 2026

Coursework: Computational Neuroscience; Computer Vision I/II; Deep Learning; Causal Inference; Mechanistic Interpretability.

Guru Gobind Singh Indraprastha University, B.Tech. Computer Science — 2019 to 2023

RESEARCH EXPERIENCE

Laboratory for Intelligent Imaging and Neural Computing (LIINC), Columbia, Graduate Researcher — Mar 2026 - Present

- Designed and ran user studies on intention decoding across multiple interaction modalities: task and stimulus design, IRB protocol support, participant recruitment, instrumentation, and synchronized multi-stream capture of EEG, gaze and behavior.
- Built out the physiological inference pipeline over multimodal neural signals: EEG preprocessing (filtering, artifact rejection, re-referencing, epoching), gaze and pupil feature extraction, temporal alignment across asynchronous streams, and decoders mapping the fused signal onto a latent representation of user intent, with cross-participant generalization as an explicit evaluation criterion.
- Integrated decoded intent with embodied foundation models so that downstream robot action selection and adaptation follow the operator's inferred goal rather than an explicit command. Targeting a complete pipeline demonstration and user study by spring 2027.

Complex Resilient Intelligent Systems Lab (CRIS), Columbia, Research Assistant — Sep 2025 - Jan 2026

- Probed learned representations across 48 transformer layers of a protein structure prediction model over 200K+ AlphaFold DB structures (contrastive representation analysis, UMAP), characterizing how residue-interaction and geometric information is organized with depth; related Predicted Aligned Error structure to interaction interfaces via ChimeraX.

RESEARCH PROJECTS

Neural intent decoding for physics-grounded humanoid control — Independent

- End-to-end pipeline mapping EEG motor imagery to continuous control of a 17-DOF humanoid in MuJoCo: compact convolutional decoder with contrastive self-supervised pretraining, intent decoder emitting joint-space action vectors, PPO/SAC refinement. Trained on PhysioNet EEGMMIDB and Schirrmeister2017; runs end to end under 100 ms, within budget for closed-loop control.

Physics-informed neural fields for body-schema acquisition — Computer Vision II

- Joint-conditioned neural radiance field for a 17-DOF humanoid, regularized by a Hamiltonian energy-conservation term computed from back-projected depth, so reconstructed geometry is mechanically as well as photometrically plausible. Paired with a motor-gated recurrent neural field and spiking-network optical flow; their mismatch flags implausible motion. Tests whether a consistent body schema emerges from visuomotor prediction alone, without body models or labels.

Computational models of neural encoding — Computational Neuroscience

- Time encoding and decoding machines (ASDM, integrate-and-fire) recovering bandlimited signals from spike timings; end-to-end auditory pathway model (cochlear filterbank, hair cell transduction, spiking populations) reconstructing stimuli from population activity; Hodgkin-Huxley and Rinzel synaptic dynamics.

Multimodal intent inference under partial observability — Independent

- Modeled user goals as latent variables in a POMDP spanning screen, VR and console interaction, maintaining a Bayesian belief state over intent and using deep Q-learning for shared autonomy, so the system assists proactively rather than on explicit command.

TECHNICAL SKILLS

Neural signal & modeling: EEG preprocessing (filtering, ICA, re-referencing, epoching), MNE-Python, band power/CSP, spiking neuron models, time encoding machines.

ML: PyTorch, TensorFlow; CNNs, transformers, RNN/LSTM; contrastive and self-supervised learning; RL (PPO, SAC, DDPG); hierarchical Bayesian modeling; causal inference.

Vision, 3D & simulation: Neural radiance fields, volume rendering, multi-view geometry, COLMAP/Open3D, camera calibration; MuJoCo, PyBullet; real-time closed-loop control, POMDPs.

Engineering: Python, TypeScript; FastAPI, React/Next.js, GraphQL; SQL, MongoDB, Redis; AWS, Linux, Git; data and experiment infrastructure.

INDUSTRY EXPERIENCE

Subconscious AI, Founding Engineer, New York — Apr 2026 to present

- Built the inference and insights layer over a hierarchical Bayesian conjoint model (AMCE estimation, kernel-density willingness-to-pay curves, demographic heterogeneity), turning noisy human choice data into decision-grade estimates; migrated the platform from R to React and FastAPI.

Outscal, Growth and Product Engineer, Delhi — Nov 2023 to Jul 2025

- Built an ML job-parsing pipeline processing 2,000+ postings daily by fine-tuning an LLM for structured extraction; designed and ran A/B experiments across the user funnel driving a 15% conversion lift.

TravClan, Frontend Engineer, Delhi — 2022 to 2023

- Internal operations platform used by 250+ staff; customer-facing booking portals serving 5,000+ agents.